

Java Swing and Menus

Feature	AWT	Swing
Full form	Abstract Window Toolkit	Swing
Package	java.awt	javax.swing
Introduced	Java 1.0	Java 1.2
Component type	Mostly heavyweight	Mostly lightweight
OS dependency	Uses native OS components	Mostly drawn/managed by Java
Look & Feel	Depends more on OS	Pluggable Look & Feel
Components	Limited/basic	Rich and extensive
Button	Button	JButton
Label	Label	JLabel
Text field	TextField	JTextField
Text area	TextArea	JTextArea
Checkbox	Checkbox	JCheckBox
Radio button	Checkbox + CheckboxGroup	JRadioButton + ButtonGroup
List	List	JList
Table	☐ No JTable equivalent	☐ JTable
Tree	☐ No JTree	☐ JTree
Tabs	☐ No direct equivalent	☐ JTabbedPane
Slider	☐ No direct equivalent	☐ JSlider
Progress bar	☐ No direct equivalent	☐ JProgressBar
Dialog	Dialog	JDialog, JOptionPane
Scroll pane	ScrollPane	JScrollPane
Customization	Less	More
Portability of appearance	Appearance may vary by OS	More consistent appearance
Suitable for	Basic GUI learning/applications	Rich desktop GUI applications

JButton

It is a Swing component that represents a button which performs an action when clicked.

```
import javax.swing.*;
```

```
public class ButtonExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JButton");

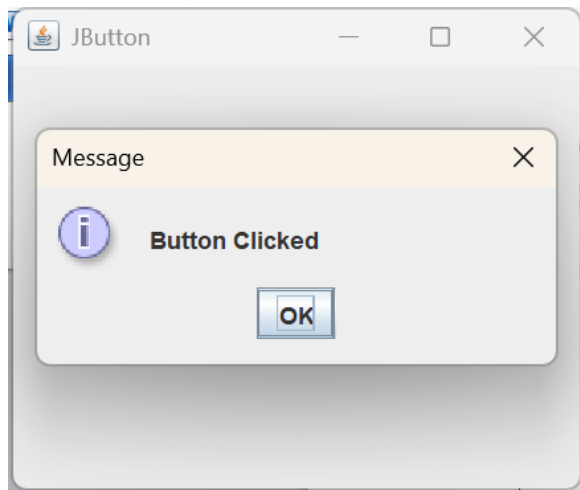
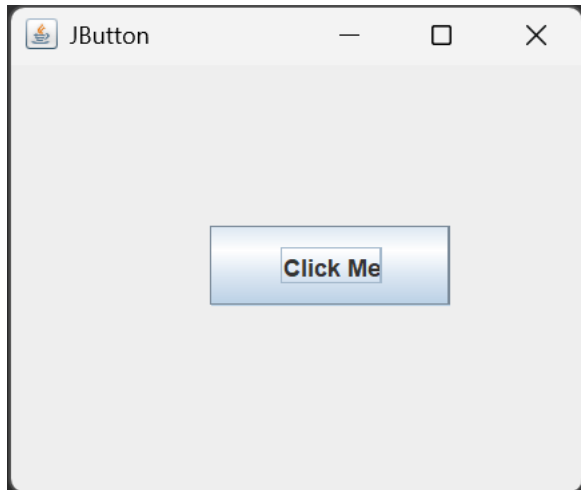
        JButton b = new JButton("Click Me");
        b.setBounds(100, 80, 120, 40);
```

```
b.addActionListener(e -> JOptionPane.showMessageDialog(f, "Button Clicked"));
// JOptionPane is a Java Swing class used to create ready-made dialog boxes.
f.add(b);
```

```

        f.setSize(300, 250);
        f.setLayout(null);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}

```



Traditional version using ActionListener

```

b.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        JOptionPane.showMessageDialog(f, "Button Clicked");
    }
});

```

The lambda expression is simply the shorter form of the above code.

JLabel

It is a Swing component used to display text, images, or both

```
import javax.swing.*;
```

```

public class LabelExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JLabel");

        JLabel l = new JLabel("Hello Java");
        l.setBounds(100, 80, 150, 40);

        f.add(l);

        f.setSize(300, 250);
        f.setLayout(null);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}

```

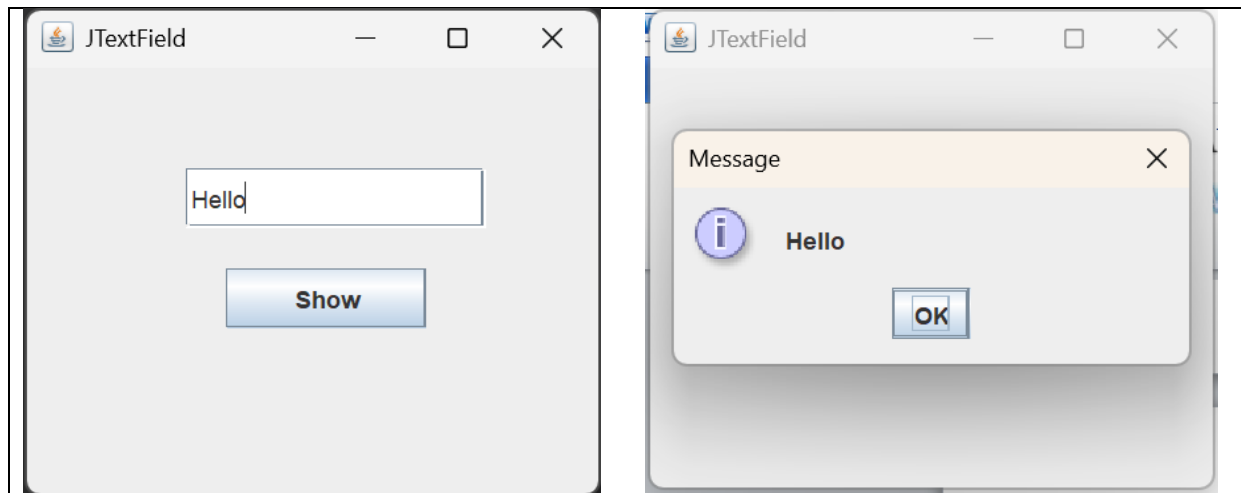


JTextField

It is a Swing component that allows the user to enter and display a single line of text.

```
import javax.swing.*;
```

```
public class TextFieldExample {  
    public static void main(String[] args) {  
        JFrame f = new JFrame("JTextField");  
  
        JTextField t = new JTextField();  
        t.setBounds(80, 50, 150, 30);  
  
        JButton b = new JButton("Show");  
        b.setBounds(100, 100, 100, 30);  
  
        b.addActionListener(e -> JOptionPane.showMessageDialog(f, t.getText())  
        );  
  
        f.add(t);  
        f.add(b);  
  
        f.setSize(300, 250);  
        f.setLayout(null);  
        f.setVisible(true);  
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
    }  
}
```



JPasswordField

It is a Swing component that allows the user to enter a password while hiding the entered characters.

```
import javax.swing.*;
```

```
public class PasswordExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JPasswordField");

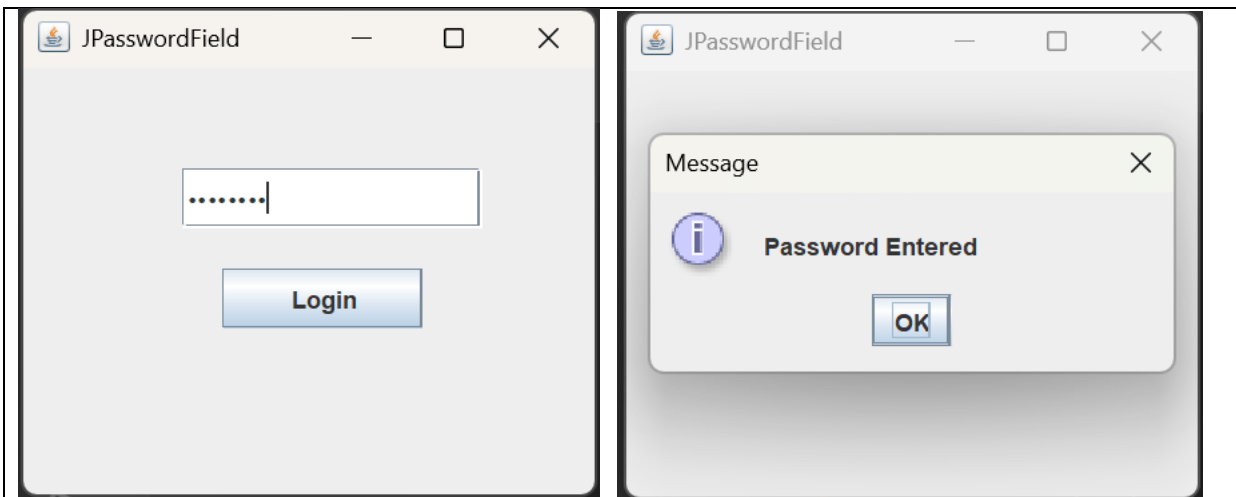
        JPasswordField p = new JPasswordField();
        p.setBounds(80, 50, 150, 30);

        JButton b = new JButton("Login");
        b.setBounds(100, 100, 100, 30);

        b.addActionListener(e->JOptionPane.showMessageDialog(f,"Password Entered")
        );

        f.add(p);
        f.add(b);

        f.setSize(300, 250);
        f.setLayout(null);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}
```



JTextArea

It is a Swing component that allows the user to enter and display multiple lines of text.

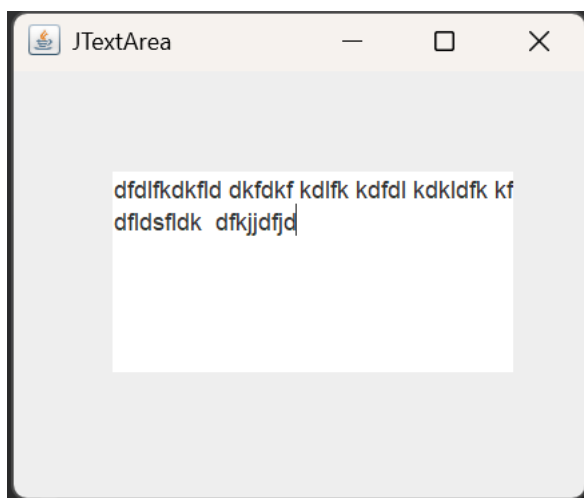
```
import javax.swing.*;
```

```
public class TextAreaExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JTextArea");

        JTextArea a = new JTextArea();
        a.setBounds(50, 50, 200, 100);

        f.add(a);

        f.setSize(300, 250);
        f.setLayout(null);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}
```



If you want scrollbars then

```
JTextArea a = new JTextArea();
JScrollPane sp = new JScrollPane(a);

f.add(sp);
```

JCheckBox

It is a Swing component that allows the user to select or deselect an option independently.

```
import javax.swing.*;
```

```
public class CheckBoxExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JCheckBox");

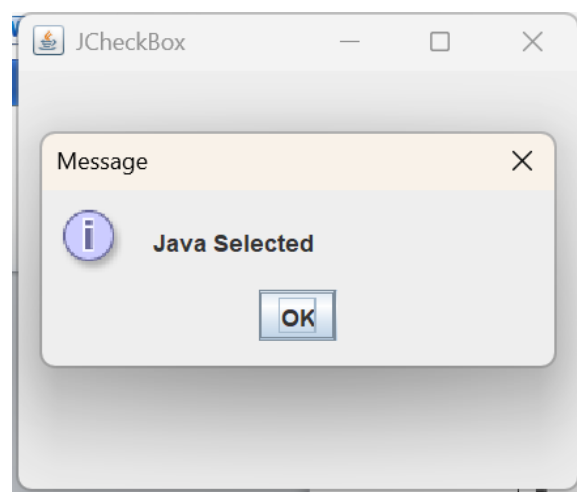
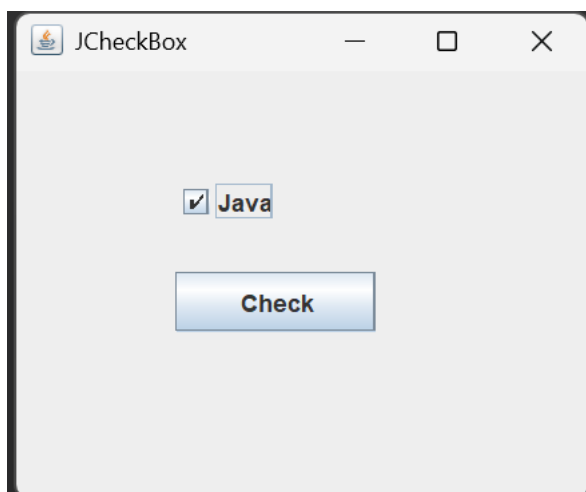
        JCheckBox c = new JCheckBox("Java");
        c.setBounds(80, 50, 100, 30);

        JButton b = new JButton("Check");
        b.setBounds(80, 100, 100, 30);

        b.addActionListener(e -> {
            if (c.isSelected())
                JOptionPane.showMessageDialog(f, "Java Selected");
            else
                JOptionPane.showMessageDialog(f, "Java Not Selected");
        });

        f.add(c);
        f.add(b);

        f.setSize(300, 250);
        f.setLayout(null);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}
```



JRadioButton

It is a Swing component that allows the user to select one option from a group of mutually exclusive options.

```
import javax.swing.*;

public class RadioButtonExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JRadioButton");

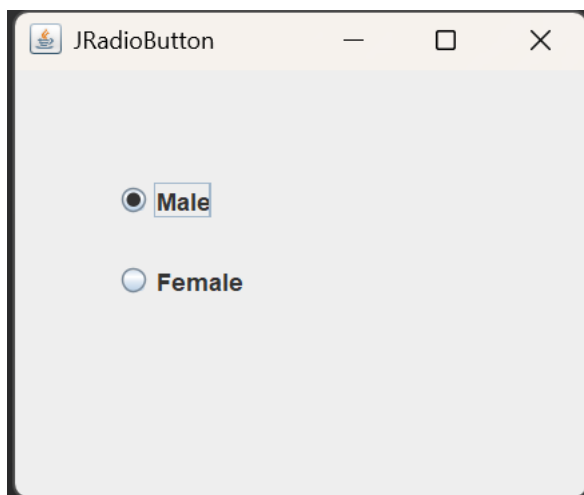
        JRadioButton r1 = new JRadioButton("Male");
        JRadioButton r2 = new JRadioButton("Female");

        r1.setBounds(50, 50, 100, 30);
        r2.setBounds(50, 90, 100, 30);

        ButtonGroup g = new ButtonGroup();
        // only one radio button can be selected at a time
        g.add(r1);
        g.add(r2);

        f.add(r1);
        f.add(r2);

        f.setSize(300, 250);
        f.setLayout(null);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}
```



JComboBox

It is a Swing component that displays a drop-down list from which the user can select an item.

```
import javax.swing.*;
```

```

public class ComboBoxExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JComboBox");

        String[] courses = {"Java", "Python", "C++"};

        JComboBox<String> c = new JComboBox<>(courses);
        c.setBounds(80, 50, 150, 30);

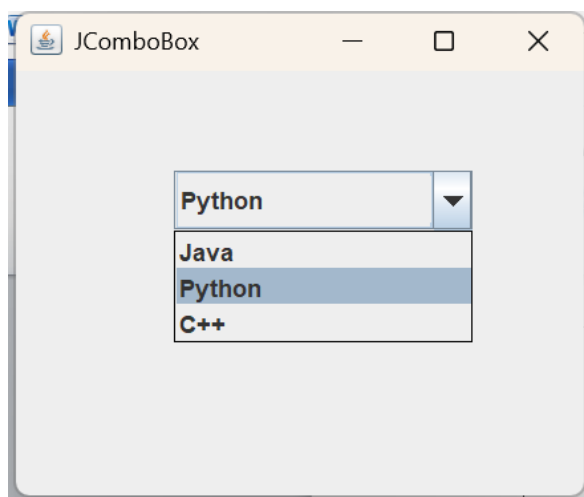
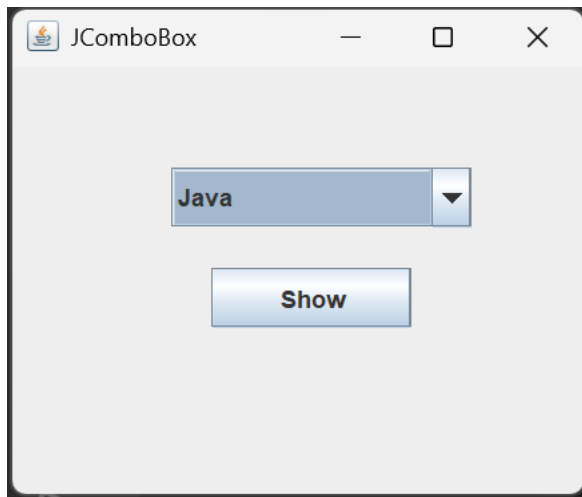
        JButton b = new JButton("Show");
        b.setBounds(100, 100, 100, 30);

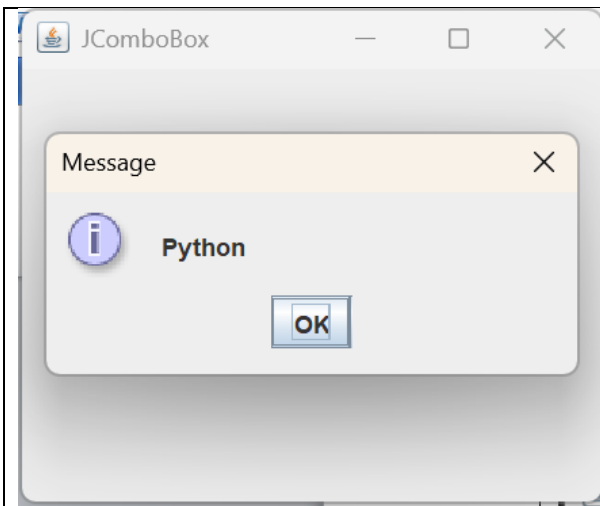
        b.addActionListener(e ->
            JOptionPane.showMessageDialog(f, c.getSelectedItem())
        );

        f.add(c);
        f.add(b);

        f.setSize(300, 250);
        f.setLayout(null);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}

```





JTable

It is a Swing component that displays data in rows and columns, similar to a table.

```
import javax.swing.*;

public class TableExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JTable");

        String[][] data = {
            {"101", "Arun"},
            {"102", "Ravi"},
            {"103", "Kumar"}
        };

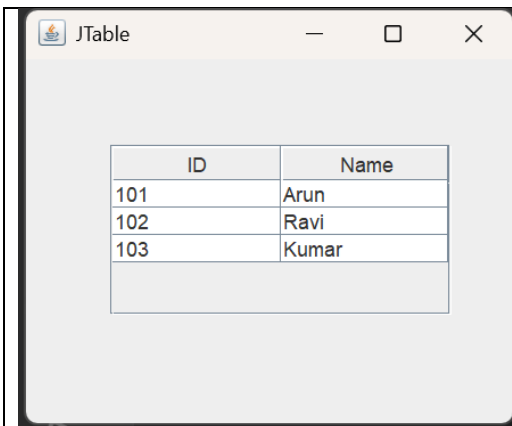
        String[] columns = {"ID", "Name"};

        JTable table = new JTable(data, columns);

        JScrollPane sp = new JScrollPane(table);
        sp.setBounds(50, 50, 200, 100);

        f.add(sp);

        f.setSize(300, 250);
        f.setLayout(null);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}
```

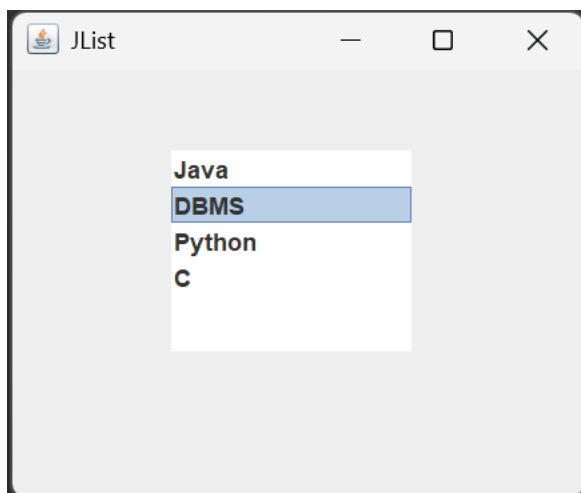


JList

It is a Swing component that displays a list of items from which the user can select one or more items.

```
import javax.swing.*;
```

```
public class ListExample {  
    public static void main(String[] args) {  
        JFrame f = new JFrame("JList");  
  
        String[] subjects = {"Java", "DBMS", "Python", "C"};  
  
        JList<String> list = new JList<>(subjects);  
        list.setBounds(80, 40, 120, 100);  
  
        f.add(list);  
  
        f.setSize(300, 250);  
        f.setLayout(null);  
        f.setVisible(true);  
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
    }  
}
```



JOptionPane

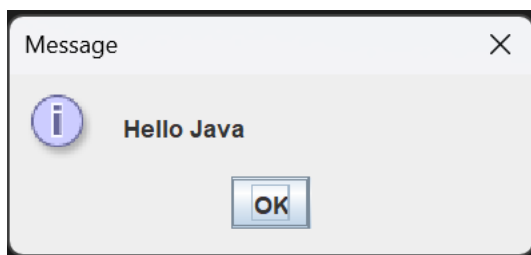
It is a Swing component that provides ready-made dialog boxes for displaying messages, getting input, and asking for confirmation.

Output Example

```
import javax.swing.*;

public class JOptionPaneExample {
    public static void main(String[] args) {

        JOptionPane.showMessageDialog(null,"Hello Java");
        // showMessageDialog(parent, message)
    }
}
```



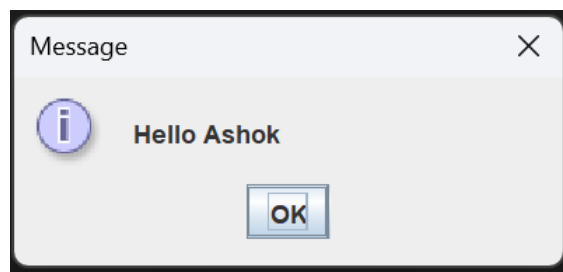
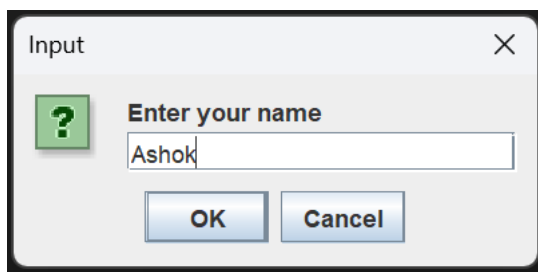
Input Example

```
import javax.swing.*;

public class JOptionPaneInput {
    public static void main(String[] args) {

        String name = JOptionPane.showInputDialog("Enter your name");
        // gets input from the user

        JOptionPane.showMessageDialog(null, "Hello " + name);
    }
}
```



JScrollBar

It is a Swing component that allows the user to scroll vertically or horizontally through content.

```
import javax.swing.*;

public class ScrollBarExample {
```

```

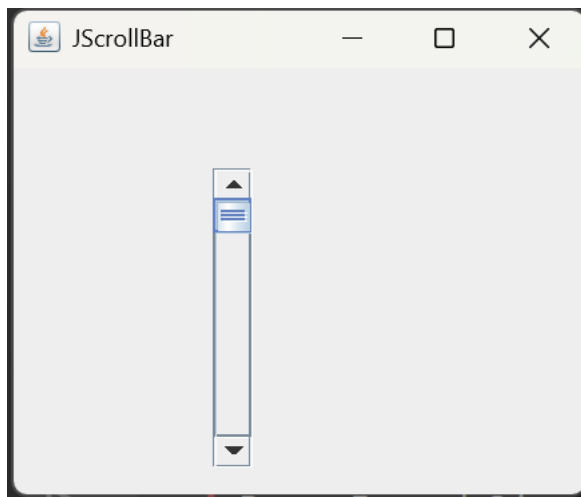
public static void main(String[] args) {
    JFrame f = new JFrame("JScrollBar");

    JScrollBar s = new JScrollBar();
    // JScrollBar s = new JScrollBar(JScrollBar.HORIZONTAL)
    s.setBounds(100, 50, 20, 150);

    f.add(s);

    f.setSize(300, 250);
    f.setLayout(null);
    f.setVisible(true);
    f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}
}

```



JMenu and JMenuItem

JMenu: It is a Swing component that represents a menu containing a list of menu items.

JMenuItem: It is a Swing component that represents an individual option or command within a menu.

```
import javax.swing.*;
```

```

public class MenuExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("Menu");

        JMenuBar bar = new JMenuBar();

        JMenu file = new JMenu("File");

        JMenuItem open = new JMenuItem("Open");
        JMenuItem exit = new JMenuItem("Exit");

        file.add(open);
    }
}

```

```

file.add(exit);

bar.add(file);

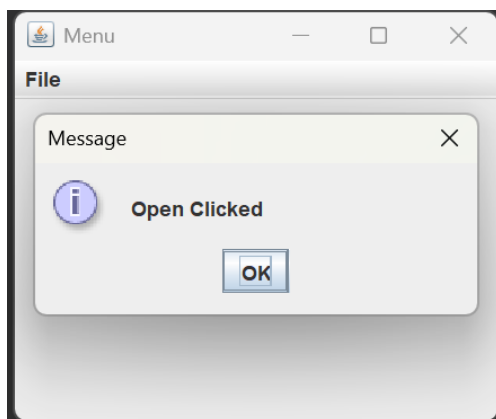
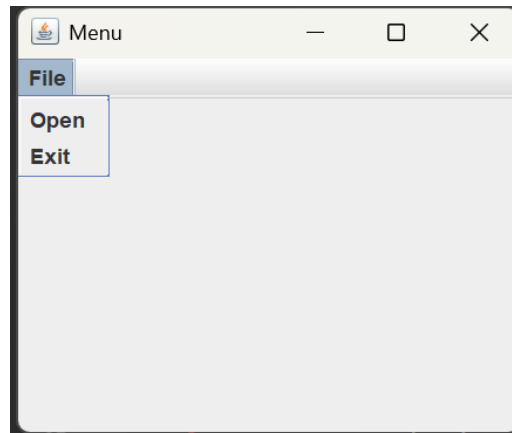
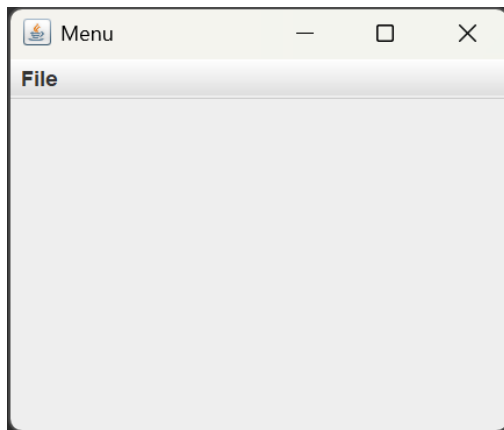
f.setJMenuBar(bar);

open.addActionListener(e ->
    JOptionPane.showMessageDialog(f, "Open Clicked")
);

exit.addActionListener(e ->System.exit(0));

f.setSize(300, 250);
f.setVisible(true);
f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}
}

```



JPopupMenu

It is a Swing component that displays a small pop-up menu containing options when triggered by the user.

```

import javax.swing.*;
import java.awt.event.*;

public class PopupExample {

```

```

public static void main(String[] args) {
    JFrame f = new JFrame("JPopupMenu");

    JTextArea a = new JTextArea();
    a.setBounds(50, 50, 200, 100);

    JPopupMenu menu = new JPopupMenu();

    JMenuItem copy = new JMenuItem("Copy");
    JMenuItem paste = new JMenuItem("Paste");

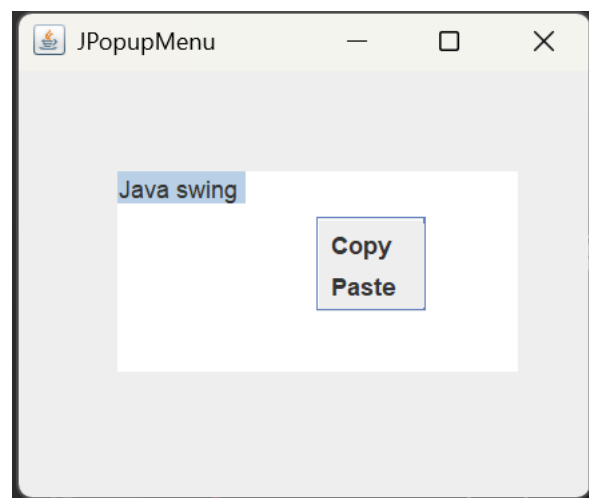
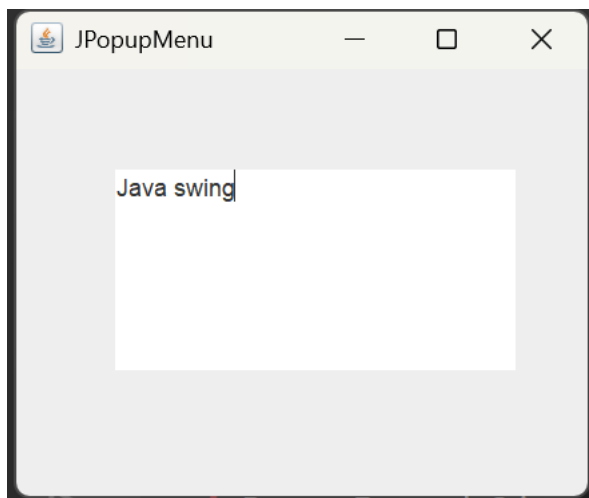
    menu.add(copy);
    menu.add(paste);

    a.setComponentPopupMenu(menu);

    f.add(a);

    f.setSize(300, 250);
    f.setLayout(null);
    f.setVisible(true);
    f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}
}

```



JCheckBoxMenuItem

It is a Swing menu component that allows the user to select or deselect an option using a check mark.

```
import javax.swing.*;
```

```

public class CheckBoxMenuExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JCheckBoxMenuItem");

        JMenuBar bar = new JMenuBar();
        JMenu view = new JMenu("View");
    }
}

```

```

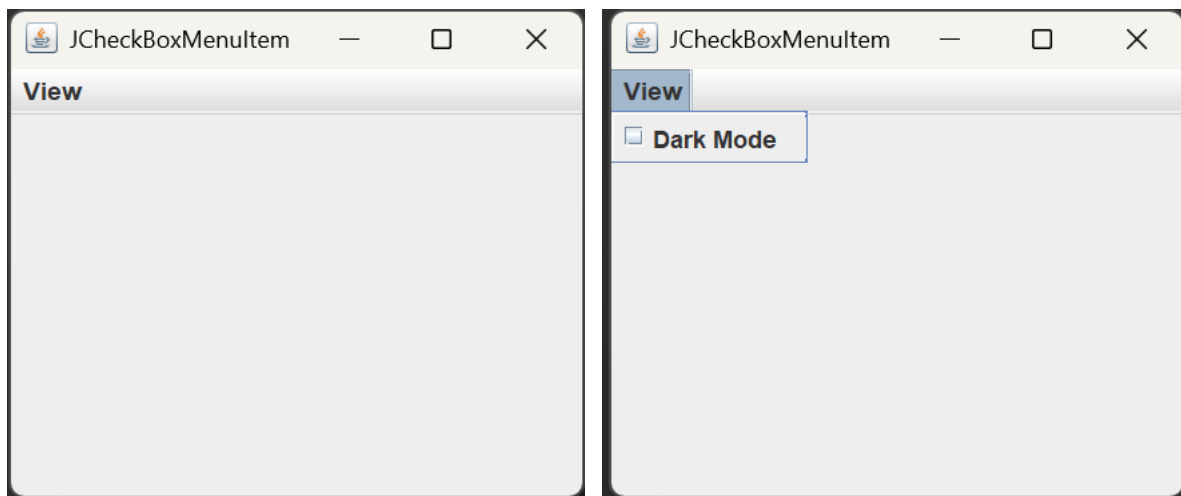
JCheckBoxMenuItem dark = new JCheckBoxMenuItem("Dark Mode");

view.add(dark);
bar.add(view);

f.setJMenuBar(bar);

f.setSize(300, 250);
f.setVisible(true);
f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}
}

```



JSeparator

It is a Swing component that provides a horizontal or vertical line to visually separate GUI components.

```

import javax.swing.*;

public class SeparatorExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JSeparator");

        JMenuBar bar = new JMenuBar();
        JMenu file = new JMenu("File");

        JMenuItem open = new JMenuItem("Open");
        JMenuItem save = new JMenuItem("Save");
        JMenuItem exit = new JMenuItem("Exit");

        file.add(open);
        file.add(save);

        file.addSeparator();

        file.add(exit);
    }
}

```

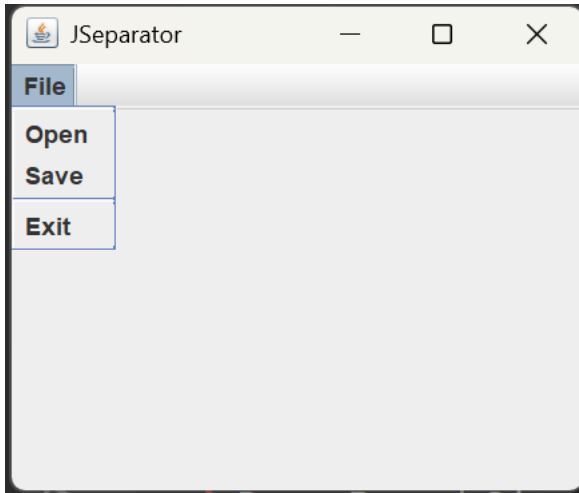
```

        bar.add(file);

        f.setJMenuBar(bar);

        f.setSize(300, 250);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}

```



JProgressBar

It is a Swing component that visually shows the progress of a task.

```

import javax.swing.*.*;

public class ProgressBarExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JProgressBar");

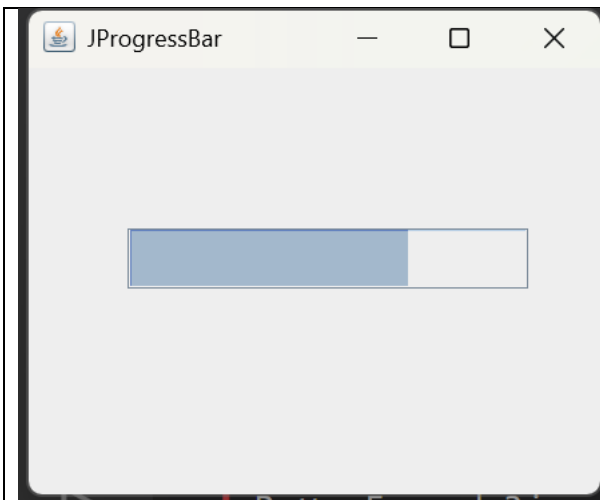
        JProgressBar p = new JProgressBar();
        p.setBounds(50, 80, 200, 30);

        p.setValue(70);

        f.add(p);

        f.setSize(300, 250);
        f.setLayout(null);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}

```

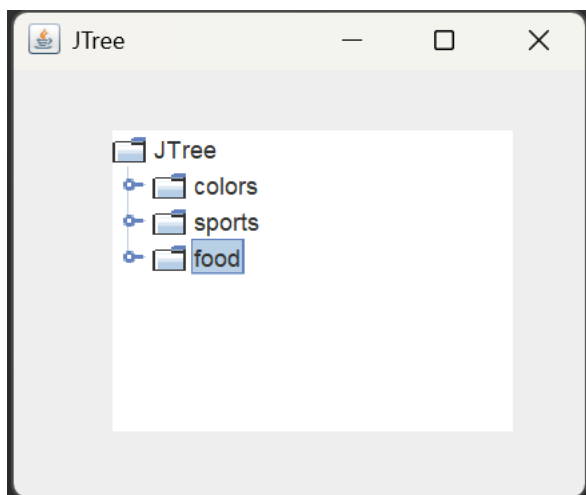



JTree

It is a Swing component that displays hierarchical data in a tree-like structure with expandable and collapsible nodes.

```
import javax.swing.*;
```

```
public class TreeExample {  
    public static void main(String[] args) {  
        JFrame f = new JFrame("JTree");  
  
        JTree tree = new JTree();  
  
        tree.setBounds(50, 30, 200, 150);  
  
        f.add(tree);  
  
        f.setSize(300, 250);  
        f.setLayout(null);  
        f.setVisible(true);  
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
    }  
}
```



Component	Main purpose
JButton	Button
JLabel	Display text
TextField	Single-line input
JPasswordField	Password input
TextArea	Multiple-line input
JCheckBox	Select/deselect
JRadioButton	Select one option
JComboBox	Drop-down list
JTable	Rows and columns
JList	List of items
JOptionPane	Dialog box
JScrollBar	Scroll bar
JMenu	Menu
JMenuItem	Item inside menu
JPopupMenu	Right-click menu
JCheckBoxMenuItem	Checkbox inside menu
JSeparator	Separating line
JProgressBar	Progress indicator
JTree	Hierarchical data

JColorChooser

It is a Swing component that allows the user to select a color from a color selection dialog.

```
import javax.swing.*;
import java.awt.*;

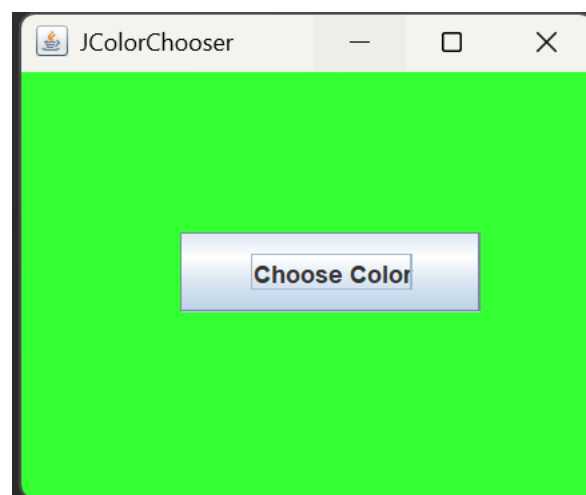
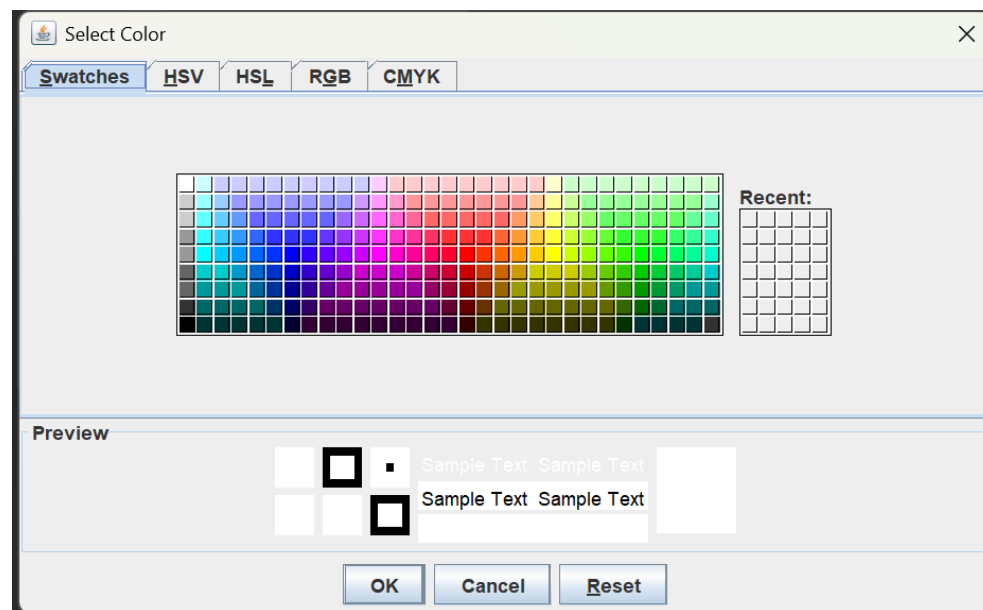
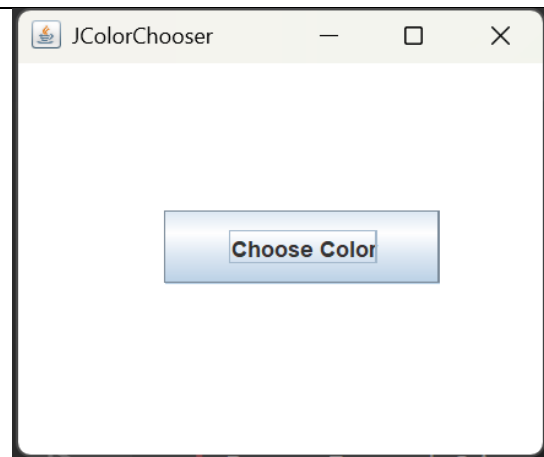
public class ColorChooserExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JColorChooser");

        JButton b = new JButton("Choose Color");
        b.setBounds(80, 80, 150, 40);

        b.addActionListener(e -> {
            Color c = JColorChooser.showDialog(f, "Select Color", Color.WHITE);
            f.getContentPane().setBackground(c);
        });

        f.add(b);

        f.setSize(300, 250);
        f.setLayout(null);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}
```



JTabbedPane

It is a Swing component that allows the user to organize multiple panels or components into separate tabs.

```
import javax.swing.*;
public class TabbedPaneExample {
```

```

public static void main(String[] args) {
    JFrame f = new JFrame("JTabbedPane");

    JTabbedPane tabs = new JTabbedPane();

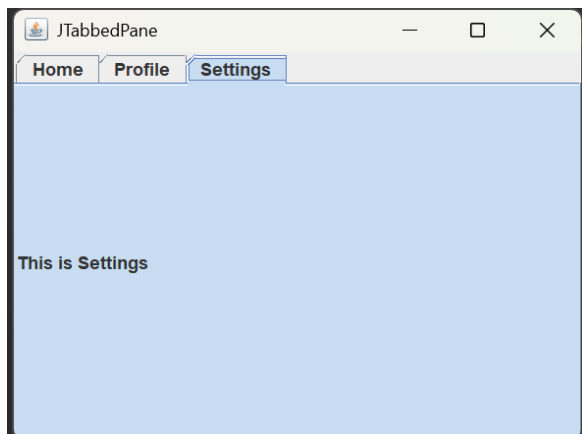
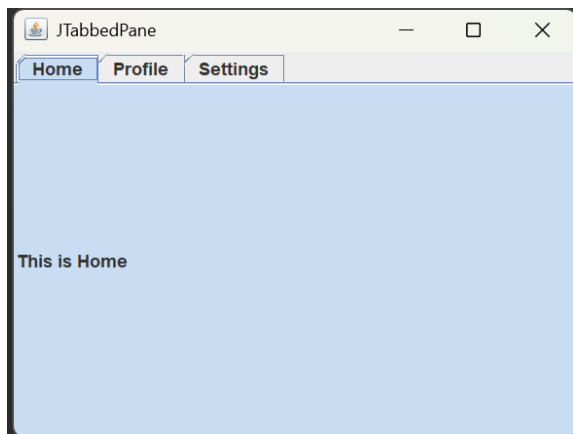
    JLabel l1 = new JLabel("This is Home");
    JLabel l2 = new JLabel("This is Profile");
    JLabel l3 = new JLabel("This is Settings");

    tabs.addTab("Home", l1); // Create a tab called Home and put component l1 inside that tab.
    tabs.addTab("Profile", l2);
    tabs.addTab("Settings", l3);

    f.add(tabs);

    f.setSize(400, 300);
    f.setVisible(true);
    f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}
}

```



JSlider

It is a Swing component that allows the user to select a value by moving a slider knob along a range.

```

import javax.swing.*;

public class SliderExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JSlider");

        JSlider s = new JSlider(0, 100, 50);
        s.setBounds(50, 70, 250, 40);

        f.add(s);

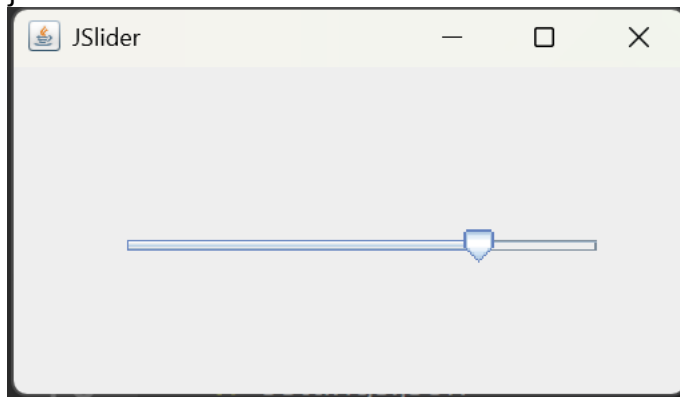
        f.setSize(350, 200);
        f.setLayout(null);
    }
}

```

```

        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}

```



JSpinner

It is a Swing component that allows the user to select a value by clicking the up/down arrow buttons

```
import javax.swing.*;
```

```

public class SpinnerExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JSpinner");

        JSpinner s = new JSpinner();
        s.setBounds(100, 70, 100, 30);

        f.add(s);

        f.setSize(300, 200);
        f.setLayout(null);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}

```



JDialog

It is a Swing dialog window that appears separately from the main JFrame

```

import javax.swing.*;

public class DialogExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JFrame");

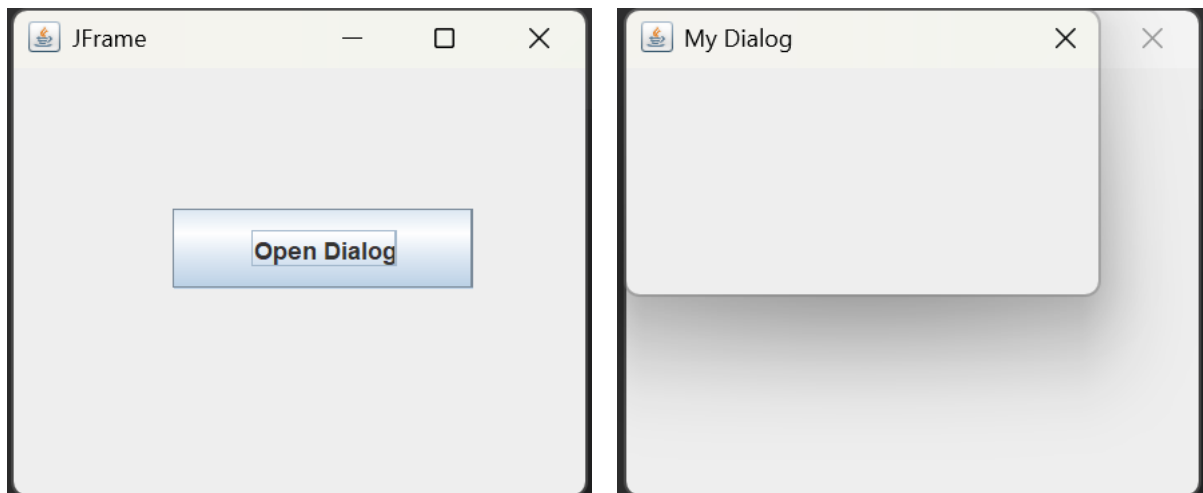
        JButton b = new JButton("Open Dialog");
        b.setBounds(80, 70, 150, 40);

        b.addActionListener(e -> {
            JDialog d = new JDialog(f, "My Dialog");
            d.setSize(250, 150);
            d.setVisible(true);
        });

        f.add(b);

        f.setSize(300, 250);
        f.setLayout(null);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}

```



JPanel

JPanel is a Swing container used to hold and organize other Swing components such as buttons, labels, text fields, etc.

```

import javax.swing.*;

public class PanelExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JPanel");

        JPanel p = new JPanel();

        JButton b = new JButton("Click");
    }
}

```

```

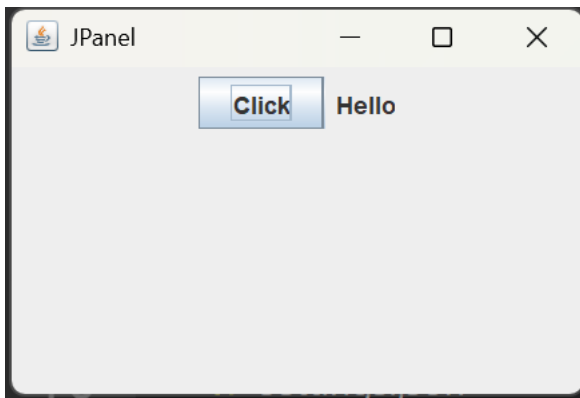
JLabel l = new JLabel("Hello");

p.add(b);
p.add(l);

f.add(p);

f.setSize(300, 200);
f.setVisible(true);
f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}
}

```



JFileChooser

It is a Swing component that allows the user to select a file or folder from the computer.

```

import javax.swing.*;

public class FileChooserExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JFileChooser");

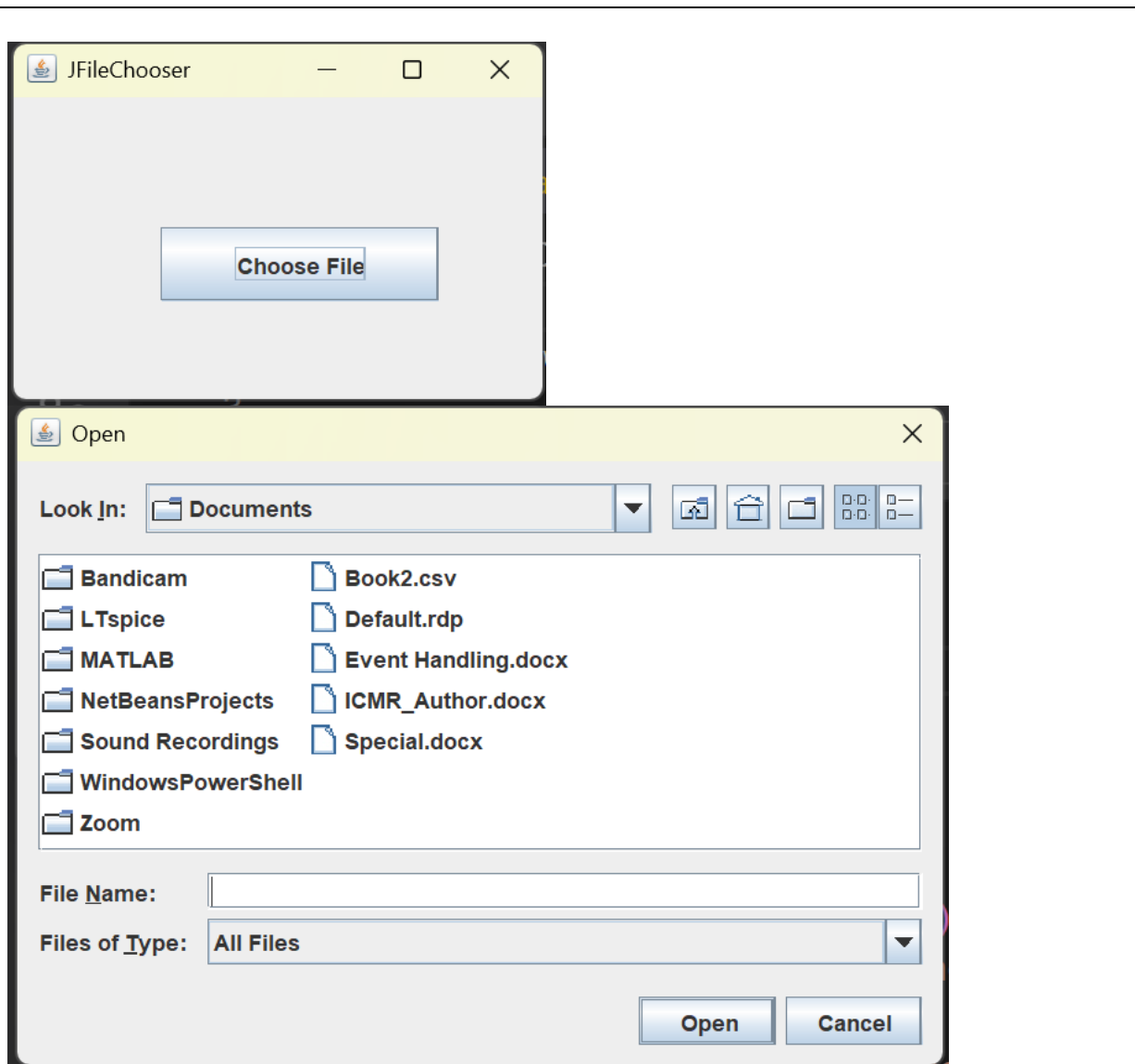
        JButton b = new JButton("Choose File");
        b.setBounds(80, 70, 150, 40);

        b.addActionListener(e -> {
            JFileChooser fc = new JFileChooser();
            fc.showOpenDialog(f); // displays the Open File dialog
        });

        f.add(b);

        f.setSize(300, 200);
        f.setLayout(null);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}

```



JToggleButton

It is a Swing component that allows the user to switch between two states, such as ON and OFF.

```
import javax.swing.*;
```

```
public class ToggleButtonExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JToggleButton");

        JToggleButton b = new JToggleButton("OFF");
        b.setBounds(80, 70, 120, 40);

        b.addActionListener(e -> {
            if (b.isSelected())
                b.setText("ON");
            else
                b.setText("OFF");
        });
    }
}
```



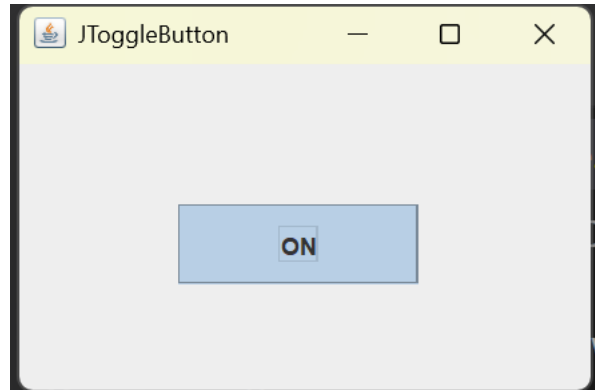
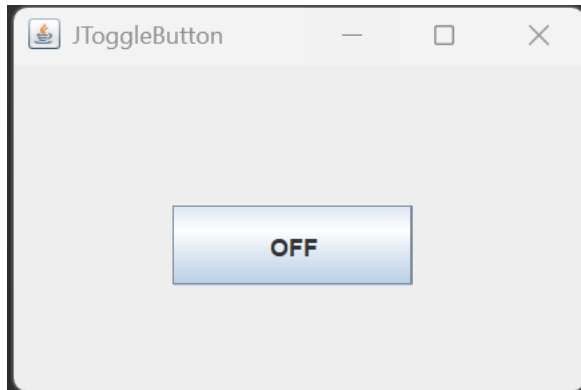
```

});

f.add(b);

f.setSize(300, 200);
f.setLayout(null);
f.setVisible(true);
f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}
}

```



JToolBar

It is a Swing component that provides a movable bar containing buttons or other controls for quick access to commands.

```

import javax.swing.*;

public class ToolBarExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JToolBar");

        JToolBar bar = new JToolBar();

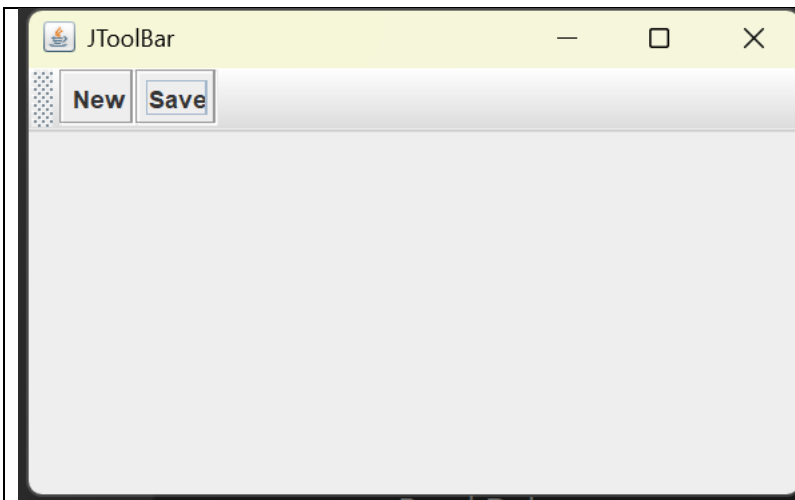
        JButton b1 = new JButton("New");
        JButton b2 = new JButton("Save");

        bar.add(b1);
        bar.add(b2);

        f.add(bar, "North");

        f.setSize(400, 250);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}

```

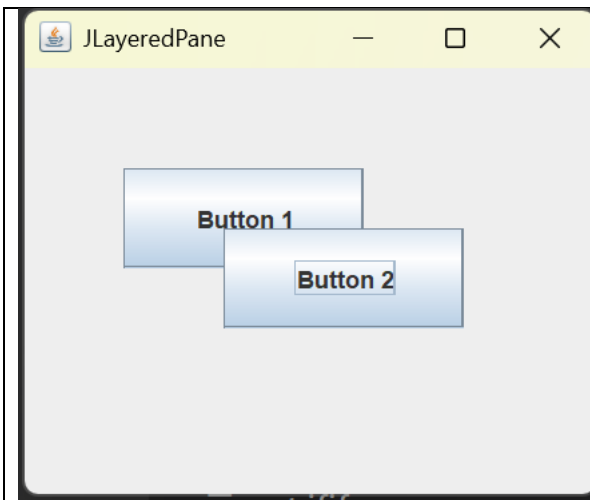


JLayeredPane

It is a Swing component that allows components to be arranged in different layers, one above another.

```
import javax.swing.*;
```

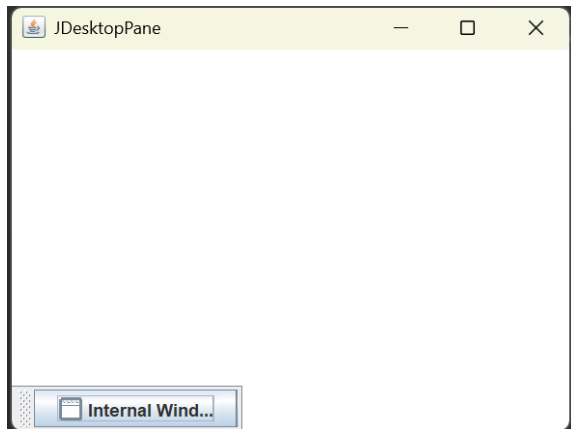
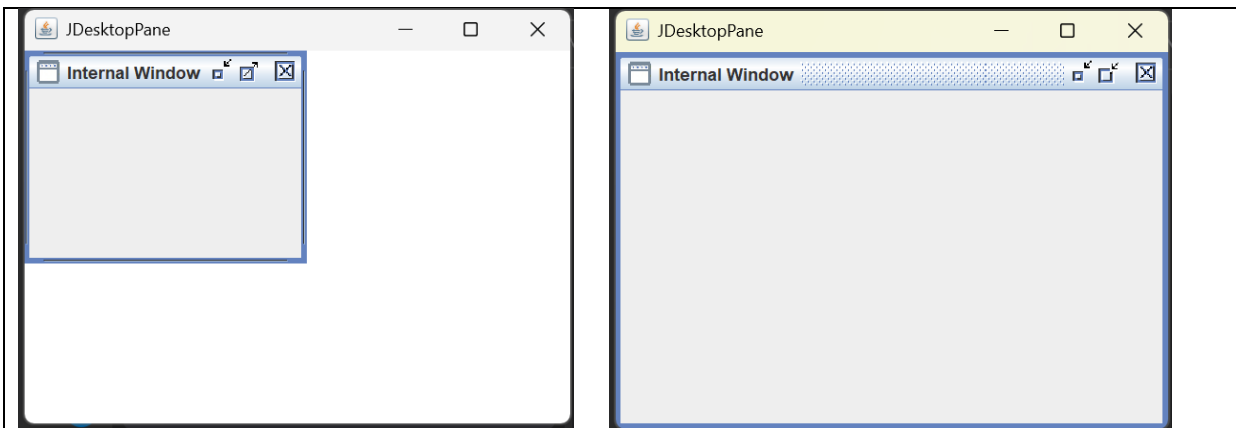
```
public class LayeredPaneExample {  
    public static void main(String[] args) {  
        JFrame f = new JFrame("JLayeredPane");  
  
        JLayeredPane lp = new JLayeredPane();  
  
        JButton b1 = new JButton("Button 1");  
        JButton b2 = new JButton("Button 2");  
  
        b1.setBounds(50, 50, 120, 50);  
        b2.setBounds(100, 80, 120, 50);  
  
        lp.add(b1, Integer.valueOf(1));  
        lp.add(b2, Integer.valueOf(2));  
  
        f.add(lp);  
  
        f.setSize(300, 250);  
        f.setVisible(true);  
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
    }  
}
```



JDesktopPane

It is a Swing component that provides a desktop area for displaying and managing multiple internal windows. import javax.swing.*;

```
public class DesktopPaneExample {  
    public static void main(String[] args) {  
        JFrame f = new JFrame("JDesktopPane");  
  
        JDesktopPane desktop = new JDesktopPane();  
  
        JFrame window = new JFrame(  
            "Internal Window",  
            true,  
            true,  
            true,  
            true  
        );  
  
        window.setSize(200, 150);  
        window.setVisible(true);  
  
        desktop.add(window);  
  
        f.add(desktop);  
  
        f.setSize(400, 300);  
        f.setVisible(true);  
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
    }  
}
```



Value	Meaning
"Internal Window"	Window title
true	Can be resized
true	Can be closed
true	Can be maximized
true	Can be minimized/iconified

JEditorPane

It is a Swing component that allows the user to display and edit text in different formats, such as plain text and HTML.

```
import javax.swing.*;
public class EditorPaneExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JEditorPane");

        JEditorPane e = new JEditorPane();

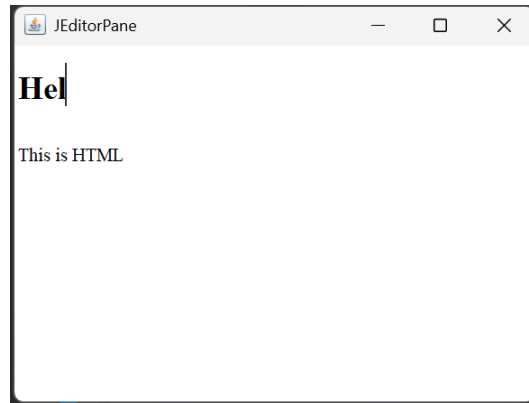
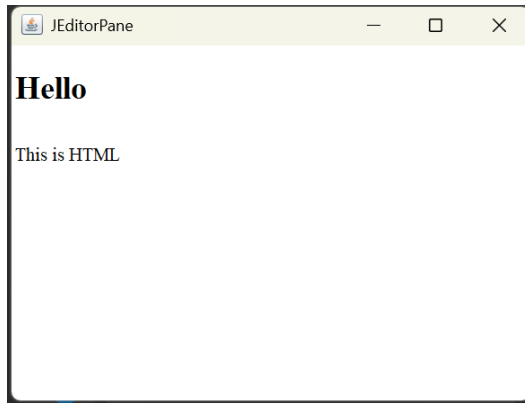
        e.setContentType("text/html");
        // If you don't specify "text/html", the HTML tags may be treated as ordinary text
        e.setText(
            "<html><h1>Hello</h1>" +
            "<p>This is HTML</p></html>"
        );
        e.setEditable(true);
    }
}
```

```

        f.add(e);

        f.setSize(400, 300);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}

```



JScrollPane

It is a Swing component that allows the user to scroll through content that is larger than the available display area.

```

import javax.swing.*;

public class ScrollPaneExample {
    public static void main(String[] args) {

        JFrame f = new JFrame("JScrollPane");

        JTextArea area = new JTextArea();

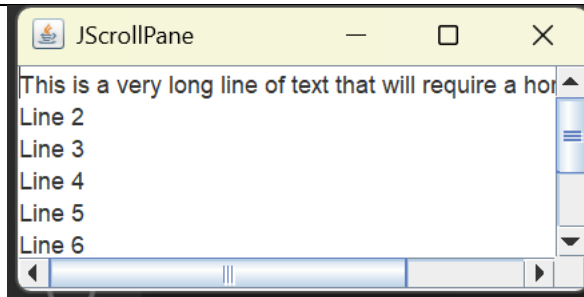
        area.setText("This is a very long line of text that will require a horizontal
scrollbar.\n"
        + "Line 2\nLine 3\nLine 4\nLine 5\nLine 6\nLine 7\nLine 8\nLine 9\nLine
10");

        JScrollPane sp = new JScrollPane(area,
        JScrollPane.VERTICAL_SCROLLBAR_ALWAYS,
        JScrollPane.HORIZONTAL_SCROLLBAR_ALWAYS);

        f.add(sp);

        f.setSize(300, 150);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}

```



JSplitPane

It is a Swing component that allows the user to divide a window into two resizable sections.

```
import javax.swing.*;

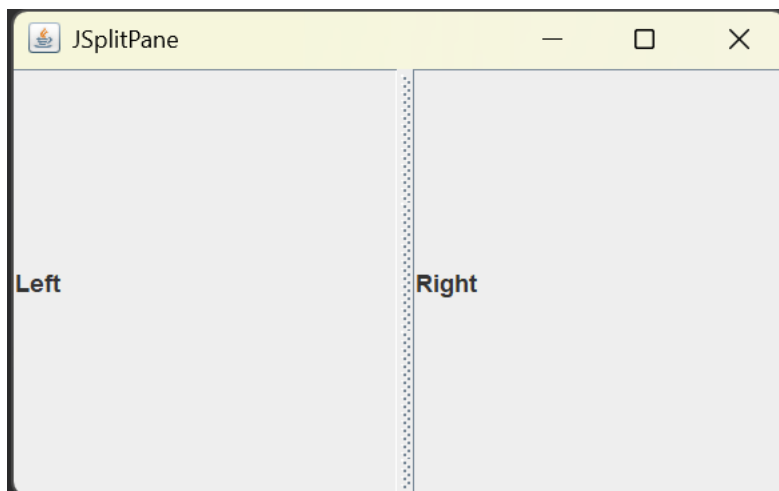
public class SplitPaneExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JSplitPane");

        JLabel l1 = new JLabel("Left");
        JLabel l2 = new JLabel("Right");

        JSplitPane split = new JSplitPane(
            JSplitPane.HORIZONTAL_SPLIT,
            l1,
            l2
        );

        f.add(split);

        f.setSize(400, 250);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}
```



JTextPane

JTextPane is a Swing component used to display and edit multi-line text with formatting.

```

import javax.swing.*;
public class TextPaneExample {
    public static void main(String[] args) {

        JFrame f = new JFrame("JTextPane Example");

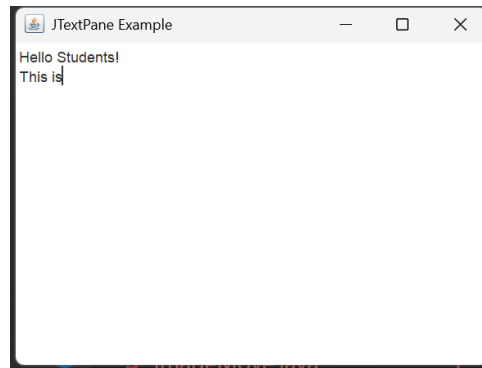
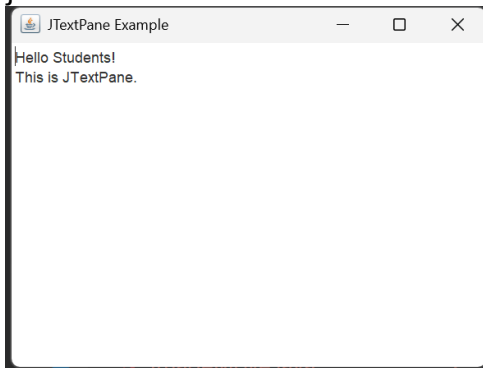
        JTextPane p = new JTextPane();

        p.setText("Hello Students!\nThis is JTextPane.");

        f.add(p);

        f.setSize(400, 300);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}

```



JRootPane

It is a Swing component that provides the main internal structure of a top-level window.

```

import javax.swing.*;
public class RootPaneExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("JRootPane");

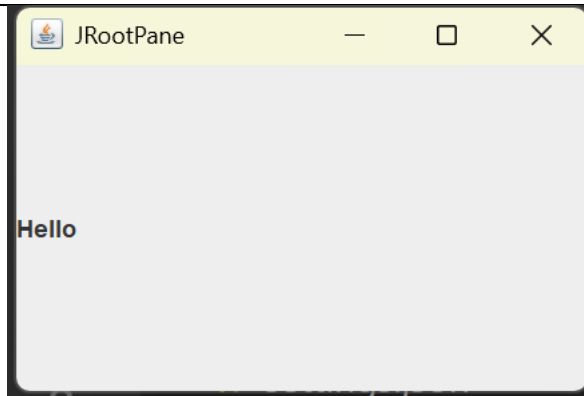
        JRootPane root = f.getRootPane();

        JLabel l = new JLabel("Hello");

        root.getContentPane().add(l);

        f.setSize(300, 200);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}

```



Using ToolTip

It is a Swing feature that displays a small information message when the user places the mouse pointer over a component.

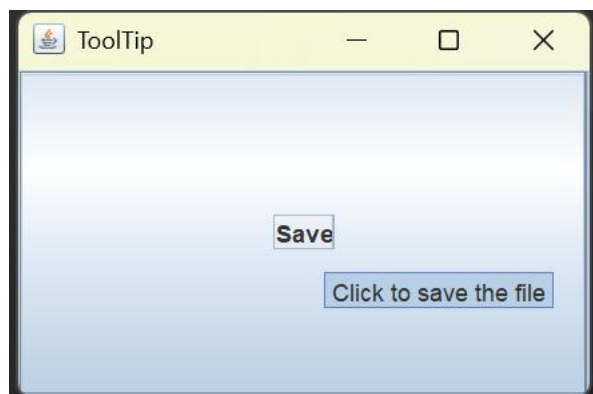
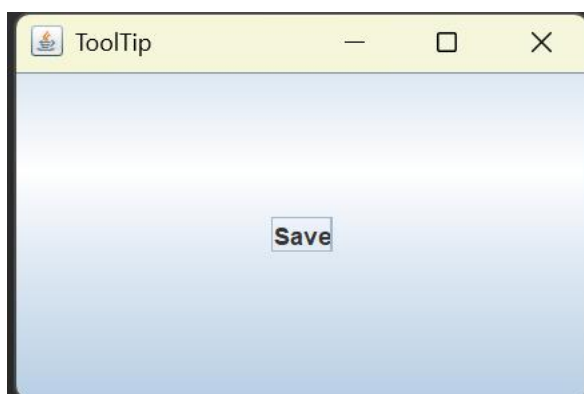
```
import javax.swing.*;
public class ToolTipExample {
    public static void main(String[] args) {
        JFrame f = new JFrame("ToolTip");

        JButton b = new JButton("Save");

        b.setToolTipText("Click to save the file");

        f.add(b);

        f.setSize(300, 200);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}
```



Change JFrame Icon

```
import javax.swing.*;
public class IconExample {
    public static void main(String[] args) {

        JFrame f = new JFrame("My Application");

        ImageIcon icon = new ImageIcon("flower.png");
```

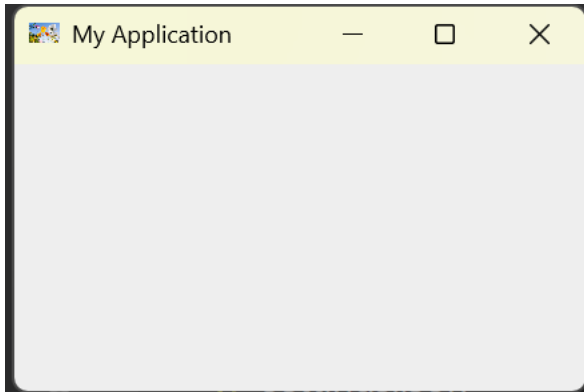


```

    f.setIconImage(icon.getImage());

    f.setSize(300, 200);
    f.setVisible(true);
    f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}
}

```



Digital Watch

```

import javax.swing.*;
import java.text.SimpleDateFormat;
import java.util.Date;

public class DigitalWatch {
    public static void main(String[] args) {

        JFrame f = new JFrame("Digital Watch");

        JLabel l = new JLabel();
        l.setBounds(80, 60, 200, 50);

        Timer t = new Timer(1000, e -> {
            Date d = new Date(); // Sun Sep 06 14:30:25 IST 2026

            SimpleDateFormat s =
                new SimpleDateFormat("HH:mm:ss");

            l.setText(s.format(d));
        });

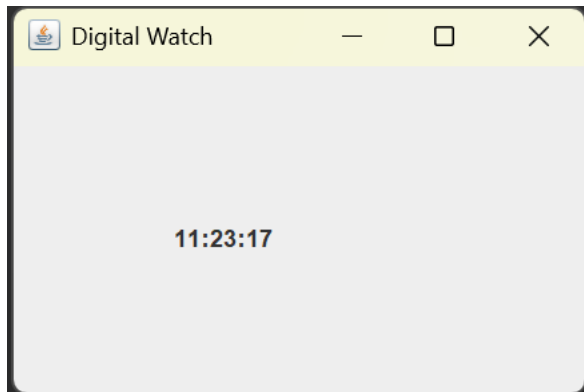
        t.start();

        f.add(l);

        f.setSize(300, 200);
        f.setLayout(null);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}

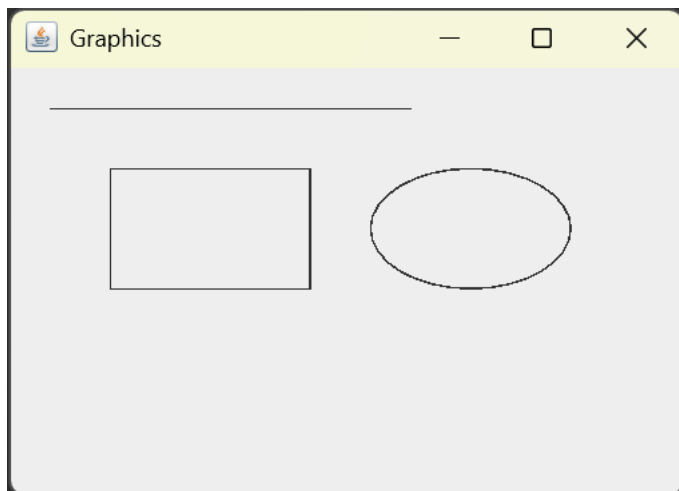
```

```
}  
}
```



Graphics in Swing

```
import java.awt.*;  
import javax.swing.*;  
  
public class GraphicsExample extends Canvas {  
  
    public void paint(Graphics g) {  
        g.drawLine(20, 20, 200, 20);  
        g.drawRect(50, 50, 100, 60);  
        g.drawOval(180, 50, 100, 60);  
    }  
  
    public static void main(String[] args) {  
        JFrame f = new JFrame("Graphics");  
  
        f.add(new GraphicsExample());  
  
        f.setSize(350, 250);  
        f.setVisible(true);  
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
    }  
}
```



Displaying an Image

```
import javax.swing.*;

public class ImageExample {
    public static void main(String[] args) {

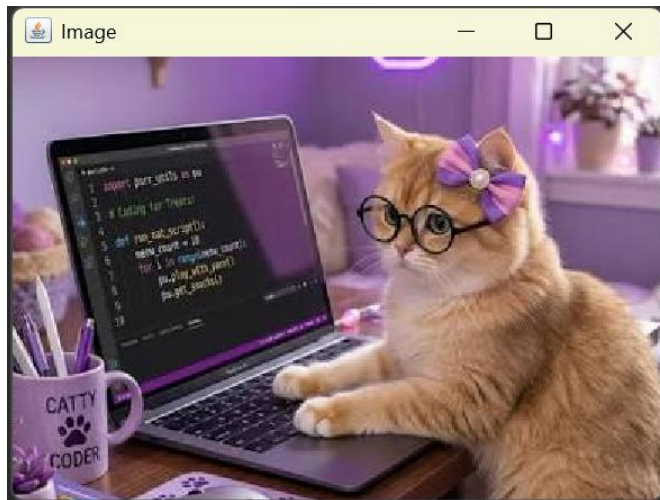
        JFrame f = new JFrame("Image");

        ImageIcon image = new ImageIcon("cat.jfif");

        JLabel l = new JLabel(image);
        // JLabel is designed to display text OR an image OR both.

        f.add(l);

        f.setSize(400, 300);
        f.setVisible(true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}
```



Component	Simple purpose
JColorChooser	Select color
JTabbedPane	Create tabs
JSlider	Select value using slider
JSpinner	Select value using arrows
JDialog	Dialog window
JPanel	Container
JFileChooser	Select file
JToggleButton	ON/OFF button
JToolBar	Toolbar
JFrame	Main window
JComponent	Base class for Swing components
JLayeredPane	Components in layers

JDesktopPane	Desktop area
JEditorPane	Display formatted/HTML content
JScrollPane	Provides scrolling
JSplitPane	Split window
JTextPane	Styled text
JRootPane	Root container of JFrame
ToolTip	Small help message
setIconImage()	Change window icon
Digital Watch	Timer + JLabel
Graphics	Draw shapes
ImageIcon	Display image